(OPTION 1) EMOTION RECOGNITION

▸AI & Design

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Topic

Emotion Recognition



Happy

VS



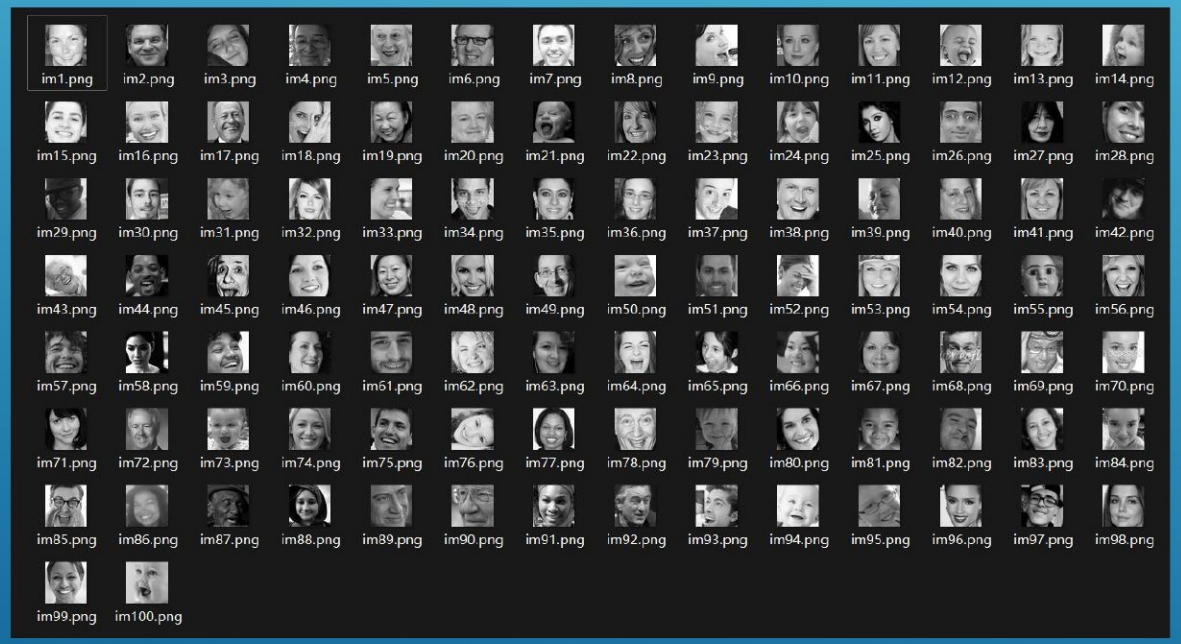
Sad

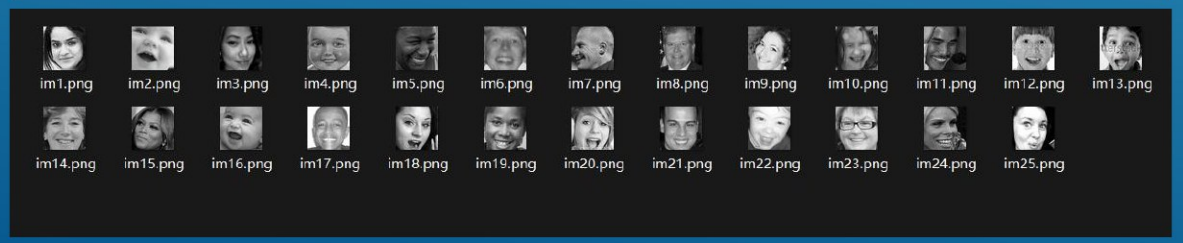
Data Preparation

Source: The project will utilize the Emotion Dataset from Kaggle, which includes labeled text samples for various emotions, including Happiness and Sadness.

Happy:

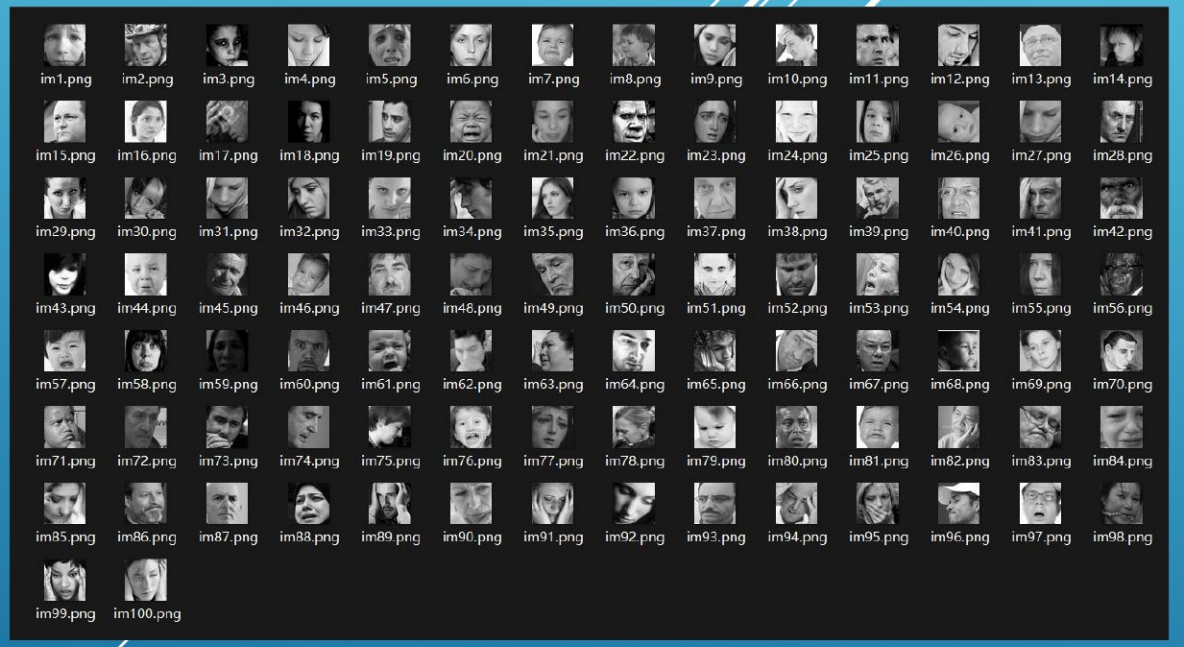
100 images for training
25 images for testing

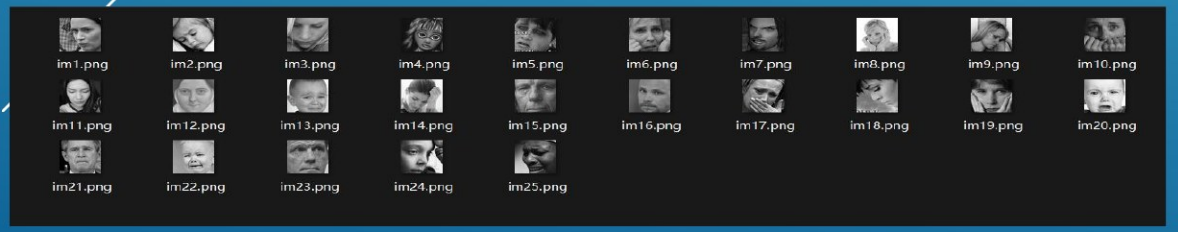




Sad:

100 images for training
25 images for testing





Training Image Recognition Model

Training Design Style Recognition Model using Colab

```
2025-01-16 16:51:56.087906: Step 220: Train accuracy = 82.0%
2025-01-16 16:51:56.087978: Step 220: Cross entropy = 0.478705
2025-01-16 16:51:56.395263: Step 220: Validation accuracy = 66.0% (N=100)
2025-01-16 16:51:59.702452: Step 230: Train accuracy = 72.0%
2025-01-16 16:51:59.702536: Step 230: Cross entropy = 0.538192
2025-01-16 16:52:00.057731: Step 230: Validation accuracy = 63.0% (N=100)
2025-01-16 16:52:03.720185: Step 240: Train accuracy = 81.0%
2025-01-16 16:52:03.720278: Step 240: Cross entropy = 0.496749
2025-01-16 16:52:04.043679: Step 240: Validation accuracy = 71.0% (N=100)
2025-01-16 16:52:07.289950: Step 250: Train accuracy = 84.0%
2025-01-16 16:52:07.290035: Step 250: Cross entropy = 0.483066
2025-01-16 16:52:07.608870: Step 250: Validation accuracy = 58.0% (N=100)
2025-01-16 16:52:10.810037: Step 260: Train accuracy = 80.0%
2025-01-16 16:52:10.810104: Step 260: Cross entropy = 0.486236
2025-01-16 16:52:11.133894: Step 260: Validation accuracy = 59.0% (N=100)
2025-01-16 16:52:14.608106: Step 270: Train accuracy = 79.0%
2025-01-16 16:52:14.608198: Step 270: Cross entropy = 0.494300
2025-01-16 16:52:14.943228: Step 270: Validation accuracy = 70.0% (N=100)
2025-01-16 16:52:18.313543: Step 280: Train accuracy = 77.0%
2025-01-16 16:52:18.313621: Step 280: Cross entropy = 0.489915
2025-01-16 16:52:18.651231: Step 280: Validation accuracy = 70.0% (N=100)
2025-01-16 16:52:21.846092: Step 290: Train accuracy = 77.0%
2025-01-16 16:52:21.846167: Step 290: Cross entropy = 0.517246
2025-01-16 16:52:22.166060: Step 290: Validation accuracy = 62.0% (N=100)
2025-01-16 16:52:25.000384: Step 299: Train accuracy = 76.0%
2025-01-16 16:52:25.000460: Step 299: Cross entropy = 0.505059
2025-01-16 16:52:25.311102: Step 299: Validation accuracy = 68.0% (N=100)
Final test accuracy = 75.0% (N=24)
```

Training Results

Train Accuracy: 76%

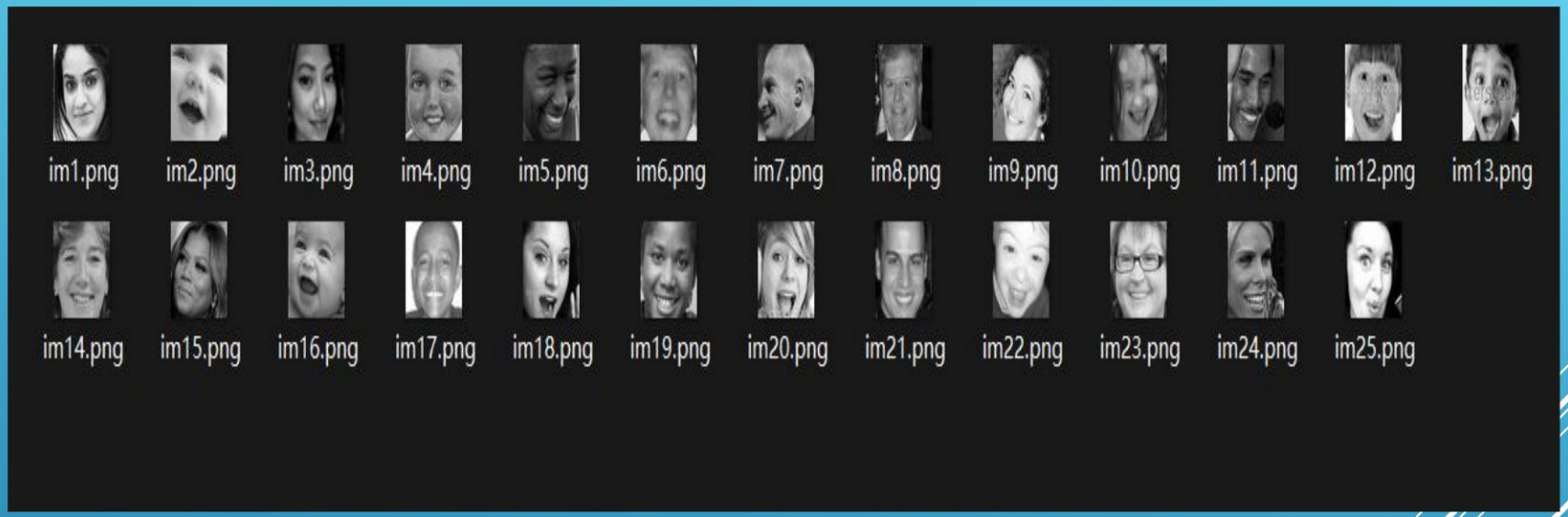
Validation Accuracy: 6% (N=100)

Final Test Accuracy: 75% (N=24)

Example of testing

Recognition Test happy Images

Image	Happy(%)	Sad(%)
im1.png	87.65	12.35
im2.png	67.87	32.13
im3.png	74.33	25.67
im4.png	76.54	23.46
im5.png	79.13	20.87
im6.png	65.43	34.57
im7.png	76.79	23.21
im8.png	78.92	21.08
im9.png	90.13	9.87
im10.png	88.76	11.24

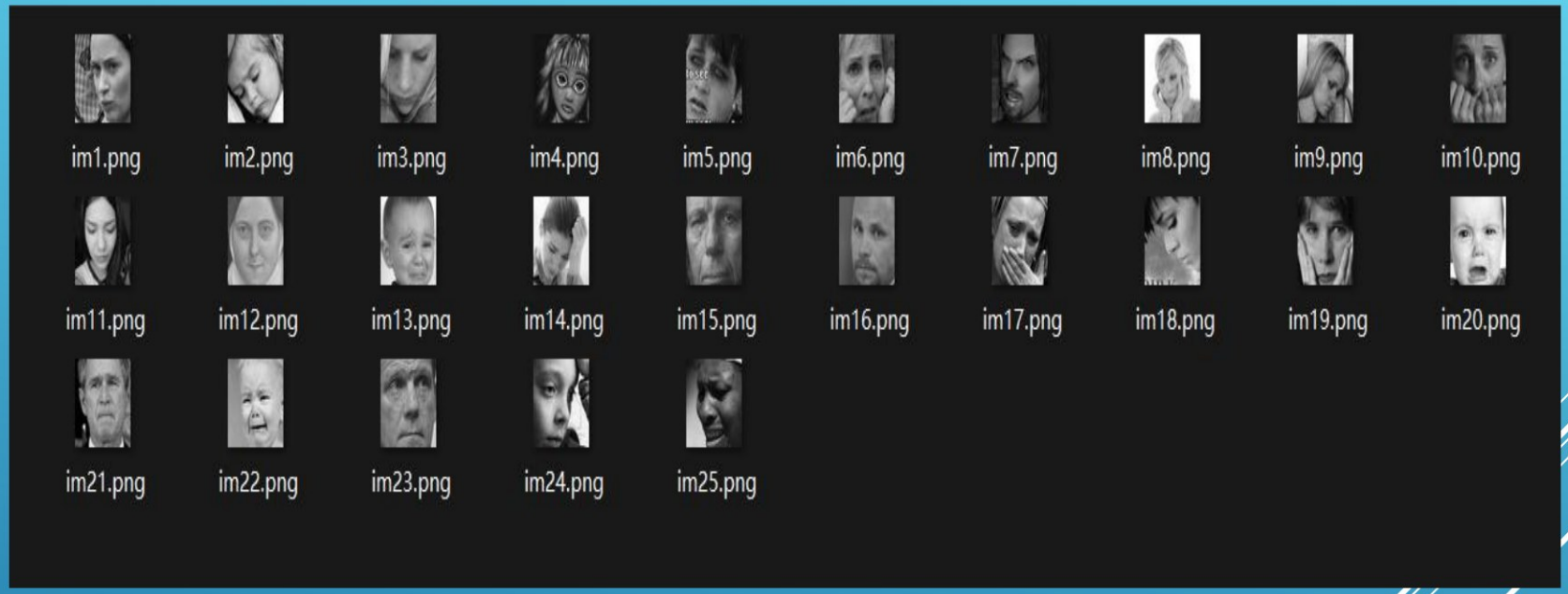


Result

Example of testing

Recognition Test sad Images

Image	Happy(%)	Sad(%)
im1.png	25.67	74.33
im2.png	15.44	84.56
im3.png	29.87	70.13
im4.png	48.79	51.21
im5.png	30.57	69.43
im6.png	19.54	80.46
im7.png	37.21	62/79
im8.png	28.65	71.35
im9.png	23.11	76.89
im10.png	46.79	53.21



Result

Reference

Training and Testing Images: Kaggle https://www.kaggle.com/datasets/ananthu017/emotion-detection-fer/data?select=train
AID_Final_Sample_Option 1.pdf
![update]tensorflow image recognition chair_vs _armchair.ipynb